The orchard-planting problem for fifteen points

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Abstract

Let $t_3(n)$ be the maximum number of lines containing exactly three points of a set of n points in the real plane. The values $t_3(n)$ are known for $n \le 14$ and for $n = 16$, and it has been known since 1974 that $t_3(15) \in \{31, 32\}$. We prove that $t_3(15) = 31$: fifteen points determine at most 31 three-point lines, so 15 is not a sporadic exception to the Burr–Grünbaum–Sloane formula $\lfloor n(n-3)/6 \rfloor + 1$ (the known exceptions are $n = 7, 11, 16, 19$). We also confirm, by the same method, the values $t_3(13) = 22$ and $t_3(14) = 26$, which had previously only been announced from an unpublished computation. For arrangements of pseudolines (equivalently, rank-3 oriented matroids) the answer is already different at $n = 13$: there is an arrangement of 13 pseudolines with 23 triple points, so the pseudoline maximum is $t_3^{ps}(13) = 23 > 22$, and the same holds at $n = 14$, where $t_3^{ps}(14) = 27 > 26 = t_3(14)$. The proof reduces the question to the exact geometric realizability of every partial Steiner triple system on 15 points with 32 blocks whose leave is a 9-edge graph with all degrees even. We enumerate these systems up to isomorphism (7,955,290 isomorphism classes, organised by 13 leave types) and show that none has a realization in the real projective plane with exactly the prescribed collinearities, using the Pappus and Desargues theorems as combinatorial filters and exact Gröbner-basis computations for the rest. Every rejection is in fact valid over every field of characteristic 0. The computations are cross-validated by two independent methods (an unfiltered algebraic run and a SAT encoding of rank-3 oriented matroids).

1 Introduction

The orchard-planting problem asks, for n trees (points) in the plane, for the maximum number $t_3(n)$ of rows (lines) containing exactly three trees. It appears in Jackson's *Rational Amusements for Winter Evenings* (1821), was popularised by Sylvester and by Dudeney, and was studied systematically by Burr, Grünbaum and Sloane [1], who showed that

$$t_3(n) \;\ge\; \left\lfloor \frac{n(n-3)}{6} \right\rfloor + 1 \tag{1}$$

by placing the points on an irreducible cubic curve so that the collinear triples are the zero-sum triples of a cyclic group of order n (a construction going back to Sylvester), and conjectured that (1) is an equality for all but finitely many n. Green and Tao [2] proved this conjecture: $t_3(n) = \lfloor n(n-3)/6 \rfloor + 1$ for all $n \ge n_0$, where their n_0 is of "double exponential type". For small n equality in (1) fails exactly for the four sporadic values $n = 7, 11, 16, 19$ known so far, where $t_3(n)$ exceeds the formula by 1, 1, 2 and 1. The exact values $t_3(n)$ are recorded in OEIS entry A003035 [3] for $n \le 14$ (the entries for $n = 13, 14$ being due to an unpublished computation of Zhao Hui Du in 2008), and $t_3(16) = 37$ is known because the counting bound of Section 2 happens to be attained by a construction. The first open case has therefore been $n = 15$, where

(1) gives 31 and counting gives $t_3(15) \leq 32$; the problem is listed as open in the Erdős problems database (Problem #669) and on MathWorld.

Theorem 1.1. $t_3(15) = 31$. *Moreover* $t_3(13) = 22$ *and* $t_3(14) = 26$.

For an arrangement of n pseudolines in the real projective plane let t_k be the number of points where exactly k pseudolines meet, and let $t_3^{ps}(n)$ be the maximum of t_3; by projective duality $t_3^{ps}(n) \geq t_3(n)$. Equivalently, by the topological representation theorem, $t_3^{ps}(n)$ is the maximum number of 3-element rank-2 flats ("lines" with exactly three points) of a simple rank-3 oriented matroid on n elements. Bokowski and Pokora [4] studied this variant for $n = 12$.

Theorem 1.2. $t_3^{ps}(13) = 23$ *and* $t_3^{ps}(14) = 27$. *In particular the pseudoline value exceeds the straight-line value by one at both* $n = 13$ *and* $n = 14$*: there is an arrangement of* 13 *pseudolines with exactly* 23 *triple points and* 9 *double points, and one of* 14 *pseudolines with* 27 *triple points, but no such arrangements of straight lines.*

Equivalently: a set of 15 points in the real (affine or projective) plane determines at most 31 lines with exactly three points, and this is attained by Sylvester's cubic-curve construction with the group $\mathbb{Z}/15\mathbb{Z}$. In the language of the Sylvester–Gallai literature, Theorem 1.1 says that there is no configuration of 15 points with exactly 9 ordinary lines and no line containing 4 or more of the points.

The proof is a computer-assisted classification in the tradition of the classification of pseudoline arrangements with many triple points by Bokowski and Pokora [4], who treated the analogous question for 12 lines. Its logical structure is as follows.

1. A counting argument together with the Kelly–Moser bound $t_2 \geq 3n/7$ on the number of ordinary lines shows that a 15-point set with 32 three-point lines would have exactly 9 ordinary lines and no line with ≥ 4 points, so its collinearity structure would be a *partial Steiner triple system* PSTS$(15, 32)$ whose *leave* (the graph of the 9 ordinary pairs) has all degrees even (Section 2).

2. We enumerate all such systems up to isomorphism: there are exactly 7,955,290 of them, distributed over the 13 possible leave graphs (Section 3).

3. For each system we decide whether it has an *exact* realization, i.e. 15 distinct points of $\mathbb{RP}^2$ whose collinear triples are precisely the 32 blocks. Two tools are used: incidence theorems (Pappus, Desargues), which reject a system whenever it contains an "open" copy of the corresponding configuration, and an exact algebraic test which, after fixing a projective frame and constructing the points by a sequence of line intersections, computes a Gröbner basis of the ideal of the remaining incidences and saturates by the non-degeneracy conditions (Section 4). No system survives (Section 5).

Section 6 describes the consistency checks: the enumeration reproduces the classical numbers of Steiner triple systems, agrees with a brute-force search where that is feasible, and the triangle-decomposable leave types reconstruct exactly the 80 Steiner triple systems of order 15; the algebraic test without any incidence-theorem filtering reaches the same verdict on every system for $n = 13$ and on large samples for $n = 15$; and an independent SAT-based test for the existence of a rank-3 oriented matroid (equivalently, a pseudoline arrangement) with the prescribed collinearities confirms the rejections on large random samples. All programs and data are available with this paper (Section 8).

2 Reduction to partial triple systems

Throughout, a *configuration* is a finite set P of points of the real projective plane $\mathbb{RP}^2$, not all on a line, and $t_k = t_k(P)$ denotes the number of lines containing exactly k points of P. Since a line missing all points of a finite set always exists, every configuration is projectively equivalent to one in the affine plane with the same numbers t_k, so working projectively loses nothing.

Lemma 2.1. *For a configuration of n points,*

$$\sum_{k\geq 2} \tbinom{k}{2}\, t_k = \binom{n}{2}, \tag{2}$$

$$t_2 \geq 3 + \sum_{k\geq 4}(k-3)\, t_k \quad \textit{(Melchior [5])}, \tag{3}$$

$$t_2 \geq 3n/7 \quad \textit{(Kelly–Moser [6])}. \tag{4}$$

Lemma 2.2. *Let P be a configuration of 15 points with $t_3 \geq 32$. Then $t_3 = 32$, $t_2 = 9$ and $t_k = 0$ for all $k \geq 4$. Similarly, a configuration of 13 points with $t_3 \geq 23$ has $(t_3, t_2) \in \{(23, 9), (24, 6)\}$ and $t_k = 0$ for $k \geq 4$; a configuration of 14 points with $t_3 \geq 27$ has $(t_3, t_2) \in \{(27, 10), (28, 7)\}$ and $t_k = 0$ for $k \geq 4$.*

Proof. For $n = 15$, (4) gives $t_2 \geq 45/7$, i.e. $t_2 \geq 7$. If $t_3 \geq 33$ then (2) leaves at most $105 - 99 = 6$ pairs for the other lines, so $t_2 \leq 6$, a contradiction. If $t_3 = 32$ then $t_2 + 6t_4 + 10t_5 + \cdots = 9$; $t_5 \geq 1$ is impossible, and $t_4 = 1$ would force $t_2 = 3 < 7$. Hence $t_4 = t_5 = \cdots = 0$ and $t_2 = 9$. The cases $n = 13$ ($t_2 \geq 6$) and $n = 14$ ($t_2 \geq 6$) are identical computations; they are also checked mechanically by the script `reduction.py`, which lists all vectors $(t_2, t_4, t_5, \dots)$ compatible with (2)–(4). □

Let now P be a 15-point configuration with $t_3 = 32$. By Lemma 2.2 any two of its 32 three-point lines share at most one point and no other triple of points is collinear; the 32 triples therefore form a *partial Steiner triple system* $\mathcal{B}$ on the point set, denoted PSTS(15, 32), and the 9 pairs of points not covered by a block are exactly the ordinary lines. We call the graph L formed by these pairs the *leave* of $\mathcal{B}$. A point p lying on $a_3(p)$ blocks lies on $a_2(p) = 14 - 2a_3(p)$ ordinary lines, so

$$\text{every vertex of the leave has even degree.} \tag{5}$$

Conversely, every configuration whose collinear triples are exactly the blocks of a PSTS(15, 32) (and whose points are distinct) has $t_3 = 32$. This motivates the following definition.

Definition 2.3. Let $\mathcal{B}$ be a set of 3-subsets (*blocks*) of an n-set X such that two blocks share at most one point. An *exact realization* of $(X, \mathcal{B})$ over a field K is an injective map $X \to K\mathbb{P}^2$ such that a triple of points of X is collinear if and only if it is a block.

Corollary 2.4. *$t_3(15) = 32$ if and only if some* PSTS(15, 32) *whose leave satisfies (5) has an exact realization over $\mathbb{R}$. Likewise $t_3(13) \geq 23$ (resp. $t_3(14) \geq 27$) if and only if some* PSTS(13, 23) *or* PSTS(13, 24) *(resp.* PSTS(14, 27) *or* PSTS(14, 28)*) with the appropriate leave parity (all degrees even for $n = 13$, all degrees odd for $n = 14$) has an exact realization over $\mathbb{R}$.*

Table 1 lists the parameters. Note that for $n = 14$, $t_3 = 28$ is *not* excluded by Lemma 2.1: it would correspond to a leave that is a perfect matching. We include this case for completeness.

n	t_3	t_2	leave	leave types
13	23	9	9 edges, all degrees even	13
13	24	6	6 edges, all degrees even	3
14	27	10	10 edges, all degrees odd	7
14	28	7	perfect matching	1
15	32	9	9 edges, all degrees even	13

Table 1: The partial triple systems to be enumerated.

3 Enumeration of the partial triple systems

3.1 Leave types

A graph with e edges in which every vertex has even degree has at most e non-isolated vertices (each non-trivial component has at least as many edges as vertices), so the even-degree leaves with 9 edges are obtained by generating all graphs on 9 vertices with 9 edges with `geng` [8] and keeping those with all degrees even; isolated vertices are then discarded and re-added as needed. There are 13 such graphs (Table 2); the four 2-regular ones are $3C_3$, C_9, $C_3 \cup C_6$ and $C_4 \cup C_5$. For $n = 14$ every vertex has odd, hence positive, degree; the 7 leaves with 10 edges are obtained from `geng` on 14 vertices with minimum degree 1.

3.2 The search

Fix a leave L on vertex set $\{0, \dots, n-1\}$, with non-isolated vertices $0, \dots, k-1$. A PSTS with leave L is the same thing as a decomposition of the edge set of $K_n - L$ into triangles. We enumerate these decompositions by depth-first search: at each node the lexicographically smallest uncovered non-leave pair $\{u, v\}$ $(u < v)$ is selected and covered by a triple $\{u, v, w\}$ for every admissible w (both $\{u, w\}$ and $\{v, w\}$ uncovered and not in L).

The isolated vertices of L are interchangeable, and we break this symmetry by the rule

> (R) at every moment of the search, the set of isolated vertices that occur in a chosen triple is an initial segment $\{k, k+1, \dots, k+m-1\}$ of the isolated vertices.

Concretely, when the selected pair $\{u, v\}$ is covered by $\{u, v, w\}$ we only allow those w for which (R) holds afterwards.

Lemma 3.1. *Every triangle decomposition of $K_n - L$ is isomorphic, by a permutation of the isolated vertices of L, to one that is visited by the search with rule (R).*

Proof. Let D be a decomposition. Run the search on D, i.e. at each step select the smallest uncovered non-leave pair and cover it with the triple of D containing it; this produces D after $|D|$ steps. Observe first that if the selected pair $\{u, v\}$ contains an isolated vertex v that has not been used yet, then v is the smallest unused isolated vertex: for any unused isolated w_0 all pairs $\{x, w_0\}$ are uncovered and not in L, so if $w_0 < v$ the pair $\{u, w_0\}$ would be a smaller uncovered pair. (The first element u of the selected pair is never an unused isolated vertex when $k \geq 1$, since $\{0, u\}$ would be smaller.) Hence a violation of (R) can only occur through the third point w of a chosen triple: w is an unused isolated vertex while some smaller isolated vertex w_0 is also unused. Let τ be the transposition $(w\, w_0)$ and $D' = \tau(D)$. Before this step no chosen triple

contains w or w_0, and the sequence of selected pairs is the same for D and D' (no pair containing w or w_0 was ever selected, by the observation above), so the search on D' agrees with the search on D up to this step and then chooses w_0 instead of w. Thus the sequence of first-use labels of isolated vertices, compared lexicographically, is strictly smaller for D' than for D. Iterating, we reach a decomposition in the orbit of D on which the search never violates (R), i.e. one that the search visits. □

Rule (R) does not break the symmetries of L itself, so a class may be visited several times; this is harmless. Every complete decomposition found is canonically labelled with *nauty* [8], applied to the point–block incidence graph with the points and the blocks as two colour classes, and stored in a hash table; each isomorphism class is written out once. (The canonical form used is the sorted list of blocks after relabelling the points by their positions in nauty's canonical labelling; because points and blocks are in different colour classes this is a complete isomorphism invariant.) The program `psts.c` implements the search.

3.3 Results of the enumeration

Table 2 gives the number of isomorphism classes of PSTS$(15, 32)$ for each of the 13 leaves. In total there are 7,955,290 classes. For $n = 13$ there are 9,348 classes with 23 blocks and 267 with 24 blocks, and for $n = 14$ there are 314,092 classes with 27 blocks and 787 with 28 blocks (Table 3).

#	graph6	structure	degrees	classes	C	N	Dg	E	S
1	`EElw`	H6	444222	19711	16772	4	2935	0	0
2	`F`AZW`	H7b	4422222	279644	246244	255	33145	0	0
3	`F`BHw`	K7a	4422222	116605	98169	162	18274	0	0
4	`F`?Nw`	F7	6222222	24734	21182	19	3533	0	0
5	`Fo@Xw`	H7c	4422222	250132	217993	256	31883	0	0
6	`F_Azo`	H7d	4422222	100372	91008	41	9323	0	0
7	`G`?G^_`	bowtie + C3	42222222	116876	96842	160	19874	0	0
8	`G?o_n_`	H8a	42222222	1906600	1671593	2968	232039	0	0
9	`G_GO^_`	H8b	42222222	1732944	1490281	2392	240271	0	0
10	`HwCW?CB`	C3 + C3 + C3	222222222	26113	20955	133	5025	0	0
11	`HoCOPHA`	C9	222222222	2335039	1980398	16553	338088	0	0
12	`HoCOOL_`	C5 + C4	222222222	506730	432769	3533	70428	0	0
13	`HwCO?SE`	C6 + C3	222222222	539790	450892	3534	85364	0	0
		total		7955290	6835098	30010	1090182	0	0

Table 2: Leave types for $n = 15$ (9 edges, all degrees even), the number of isomorphism classes of PSTS$(15, 32)$ with that leave, and the verdicts of the realizability tests of Section 4 (C: rejected by a Pappus or Desargues closure; N: incidence equations have no complex solution; Dg: all complex solutions degenerate; E: all non-degenerate solutions carry an extra collinearity; S: survivors).

$n = 13$, 23 **blocks**

#	graph6	structure	degrees	classes	C	N	Dg	E	S
1	EElw	H6	444222	57	55	0	2	0	0
2	F`AZW	H7b	4422222	535	435	16	84	0	0
3	F`BHw	K7a	4422222	236	199	5	32	0	0
4	F`?Nw	F7	6222222	60	45	2	13	0	0
5	Fo@Xw	H7c	4422222	453	346	18	89	0	0
6	F_Azo	H7d	4422222	171	137	12	22	0	0
7	G`?G^_	bowtie + C3	42222222	157	114	10	33	0	0
8	G?o_n_	H8a	42222222	2416	1967	123	326	0	0
9	G_GO^_	H8b	42222222	2233	1818	97	318	0	0
10	HwCW?CB	C3 + C3 + C3	222222222	38	32	1	5	0	0
11	HoCOPHA	C9	222222222	2060	1633	109	318	0	0
12	HoCOOL_	C5 + C4	222222222	438	376	11	51	0	0
13	HwCO?SE	C6 + C3	222222222	494	438	11	45	0	0
		total		9348	7595	415	1338	0	0

$n = 13$, 24 **blocks** (three leave types together): 267 classes: P 238, D 3, N 12, Dg 14, E 0, S 0.

$n = 14$, 27 **blocks**

#	graph6	structure	degrees	classes	C	N	Dg	E	S
1	g6		33311111111111	19708	13427	159	6121	1	0
2	g6		33311111111111	125130	93175	805	31150	0	0
3	g6		33311111111111	117296	90542	698	26056	0	0
4	g6		53111111111111	25177	20937	226	4014	0	0
5	g6		33311111111111	17152	13381	143	3628	0	0
6	g6		53111111111111	9264	7893	58	1313	0	0
7	g6		71111111111111	365	322	6	37	0	0
		total		314092	239677	2095	72319	1	0

$n = 14$, 28 **blocks** (perfect-matching leave): 787 classes, all non-orientable as oriented matroids (hence not realizable by straight lines), which excludes $t_3(14) = 28$ and $t_3^{ps}(14) = 28$.

Table 3: The same data for $n = 13$ and $n = 14$.

4 Exact realizability

Let $\mathcal{B}$ be a PSTS on $X = \{0, \ldots, n-1\}$ with all blocks of size 3. We describe two sufficient conditions for the non-existence of an exact realization, both valid over every field of characteristic 0 (and, for the first, over every field).

4.1 Incidence-theorem closure

Let $\mathcal{P}$ denote the Pappus configuration 9_3 and $\mathcal{D}$ the Desargues configuration 10_3, each viewed as a set of 3-element "lines" on an abstract point set. Both are line-transitive: their automorphism groups have orders 108 and 120 and act transitively on the lines (this is easily checked with nauty).

Lemma 4.1. *Let $\mathcal{C} \in \{\mathcal{P}, \mathcal{D}\}$, let ℓ_0 be a line of $\mathcal{C}$ and let φ be an injective map from the points of $\mathcal{C}$ to X that maps every line of $\mathcal{C}$ other than ℓ_0 onto a block of $\mathcal{B}$. If $\varphi(\ell_0)$ is not a block of $\mathcal{B}$,*

then $\mathcal{B}$ has no exact realization over any field (in the case $\mathcal{D}$) or over any field with commutative multiplication, i.e. any field (in the case $\mathcal{P}$).

Proof. By line-transitivity we may assume that ℓ_0 is the "conclusion line" of the respective theorem. Suppose an exact realization ρ exists. For Pappus, let $\{A, B, C\}$ and $\{a, b, c\}$ be the two base lines of $\mathcal{C}$ and $X_0 = \varphi(Ab) \cap \varphi(aB)$ etc. the three cross points; all nine images are distinct points, the two base lines are realized by distinct lines (they are distinct blocks), no base point lies on the other base line (that would be a fourth point on a block), and the six cross lines are distinct blocks so their pairwise intersections are single points. Pappus' theorem then says that the three cross points are collinear. Since ρ is exact, their triple must be a block, i.e. $\varphi(\ell_0) \in \mathcal{B}$. For Desargues, the three lines through the centre are distinct blocks, the two triangles are non-degenerate (if the three vertices of a triangle were collinear in ρ they would form a block, and then the three intersection points with the corresponding sides of the other triangle would lie on the same line, giving six points on a block), corresponding sides are distinct blocks, so Desargues' theorem applies and the same argument shows $\varphi(\ell_0) \in \mathcal{B}$. □

The lemma is applied exhaustively: for Pappus, over all ordered pairs of disjoint blocks and all 3! bijections between them; for Desargues, over all points o, all triples of blocks through o and all 4 inequivalent orientations of the corresponding perspective triangles. This is implemented in `closure_fast.py`; a slower generic implementation that searches for injective embeddings of $\mathcal{C} \setminus \ell_0$ directly (`realize.py`) agrees with it on all systems tested.

4.2 The algebraic test

Let $\mathcal{B}$ be a system that passes the closure filter. Choose four points of X, no three in a block (a *frame*). In any exact realization the frame is in general position, so after a projective transformation its points are (1:0:0), (0:1:0), (0:0:1), (1:1:1). We then place the remaining points one at a time in a greedy order, maintaining polynomial homogeneous coordinates in a polynomial ring $\mathbb{Q}[t_1, \ldots, t_m]$:

- if the next point p lies on at least two blocks whose other two points are already placed, p is the intersection of the corresponding two lines, computed by cross products (*determined* point);
- if p lies on exactly one such block $\{a_1, a_2, p\}$, we write $p = a_1 + t\, a_2$ with a new parameter t;
- otherwise we write $p = (1{:}t{:}t')$ with two new parameters.

Among all frames (with an early exit once at most two parameters are needed) we take the order with the fewest parameters; for the systems in this paper $m \leq 2$ always.

Lemma 4.2. *If ρ is an exact realization over a field K of characteristic 0, normalised on the frame, then there is $\tau \in K^m$ such that every constructed coordinate vector, evaluated at τ, is a non-zero multiple of the corresponding point of ρ.*

Proof. Induction along the construction. For a determined point the two lines are distinct in ρ (otherwise four points would be collinear), so their cross product is a non-zero multiple of $\rho(p)$. For a point on one known line, $\rho(p) \neq \rho(a_2)$ so $\rho(p) = a_1 + t a_2$ for a unique $t \in K$. For a free point, no block contains p and two placed points; in particular $\{e_2, e_3, p\}$, where e_2, e_3 are the

second and third frame points, is not a block, so by exactness $\rho(p)$ is not on the line $x = 0$ and can be normalised to $(1{:}t{:}t')$. □

Let $I \subset \mathbb{Q}[t_1, \dots, t_m]$ be the ideal generated by the 3×3 determinants of the coordinate vectors of all blocks (including those used in the construction; the latter vanish identically). By Lemma 4.2, an exact realization yields a point $\tau \in V(I)$ at which, in addition, the following polynomials do not all vanish: for every determined point p, its three coordinates (*well-definedness*); for every pair of points, the three 2×2 minors of their coordinate vectors (*distinctness*); and for every non-block triple, its determinant (*exactness*). Recall that for an ideal J, $\operatorname{sat}(I, J) = \bigcup_k (I : J^k)$ and $V(\operatorname{sat}(I, J))$ is the Zariski closure of $V(I) \setminus V(J)$; in particular $\operatorname{sat}(I, J) = (1)$ if and only if $V(I) \subseteq V(J)$ over $\overline{\mathbb{Q}}$, and then also over every field of characteristic 0. The test proceeds in three stages:

N compute a Gröbner basis of I; if it is $\{1\}$, there is no solution at all (*no complex solution*);

D otherwise saturate successively by the well-definedness and distinctness ideals; if the result is (1), every solution is degenerate (*degenerate*);

E otherwise saturate successively by the principal ideals of the non-block determinants; if the result is (1), every non-degenerate solution has an extra collinearity (*extra collinearity*).

If a system survives all three stages, the remaining ideal describes the Zariski closure of the set of exact realizations (up to projective equivalence) and its real points must be analysed further; this did not happen for any system in this paper.

Lemma 4.3. *If a system receives verdict N, D or E, it has no exact realization over any field of characteristic* 0*.*

Proof. Suppose ρ is an exact realization over K and let τ be as in Lemma 4.2. Then $\tau \in V(I)$ and $\tau \notin V(J)$ for every saturation ideal J used. If $f \in \operatorname{sat}(I, J)$ then $fg^k \in I$ for some $g \in J$-products with $g(\tau) \neq 0$ (more precisely, $fJ^k \subseteq I$ and some generator product does not vanish at τ), hence $f(\tau) = 0$. By induction τ is a zero of every successive saturation, which is impossible if the final ideal is (1). □

The computations are carried out in Singular [9], with all arithmetic over $\mathbb{Q}$. We note one implementation pitfall: in the Singular version used, the expression `sat(I,J)[1]` returns only the *first generator* of the saturated ideal rather than the ideal; the list returned by `sat` must be assigned to a variable first. Since saturation is monotone, the erroneous idiom can only produce spurious *survivors*, never spurious rejections; all results reported here were computed with the corrected code.

4.3 Oriented matroids

A third, independent test is used for verification. An exact realization of $\mathcal{B}$ over $\mathbb{R}$ induces a rank-3 chirotope χ on X (the sign of the determinant of each ordered triple) whose zero set is exactly the set of blocks and which satisfies, for every $a \in X$ and every four further points $b < c < d < e$, the three-term Grassmann–Plücker condition: the three numbers $\chi(abc)\chi(ade)$, $-\chi(abd)\chi(ace)$, $\chi(abe)\chi(acd)$ are either all zero or include both a positive and a negative one. The same holds for the chirotope of an arrangement of pseudolines with the prescribed multiple points (Folkman–Lawrence). We encode the existence of such a χ as a satisfiability problem with

one Boolean variable per non-block triple (`omcnf.c`), break the reorientation symmetry by fixing one triple containing each point, and solve with CaDiCaL [10]. Unsatisfiability certifies that no real configuration—and no pseudoline arrangement—has the given collinearity structure. Since the underlying matroid of a partial linear space is a matroid, the three-term relations characterise chirotopes here [11, Thm. 3.6.2], so satisfiability conversely produces a pseudoline arrangement.

5 Results

Tables 2 and 3 record, for each leave type, the number of isomorphism classes and how many received each verdict. In every case the verdicts N, D, E and the closure rejections account for all classes; no class survives. For $n = 14$ the value $t_3 = 28$ is handled together with the pseudoline computation: all 787 classes of PSTS$(14, 28)$ are non-orientable, hence not realizable by straight lines either, so $t_3(14) \leq 27$; combined with the rejection of all PSTS$(14, 27)$ this gives $t_3(14) = 26$. By Corollary 2.4 the $n = 15$ and $n = 13$ computations prove Theorem 1.1. Of the 7,955,290 classes, 6,835,098 are rejected by an incidence theorem, 1,090,182 because every complex solution of the incidence equations is degenerate, and 30,010 because those equations have no complex solution at all.

Remark 5.1. All rejections are valid over every field of characteristic 0. Hence no set of 15 points of $\mathbb{CP}^2$ has exactly 9 ordinary lines, 32 three-point lines and no four-point line either; note however that the reduction of Lemma 2.2 uses the Kelly–Moser inequality, which is specific to the real plane, so this does not by itself determine the complex analogue of $t_3(15)$.

5.1 Pseudolines

The reduction of Section 2 uses only the counting identity, Melchior's inequality and the Kelly–Moser (equivalently Csima–Sawyer) bound on ordinary points, all of which hold for arrangements of pseudolines; so a pseudoline arrangement with the maximum number of triple points and no point of higher multiplicity again has the combinatorial type of one of the partial triple systems enumerated above, and $t_3^{ps}(n)$ is decided by the *orientability* of these systems (a rank-3 oriented matroid is exactly a pseudoline arrangement, by the topological representation theorem). Orientability is tested with the SAT encoding of Section 4.3, which is not restricted to straight lines.

For $n = 13$: the counting bound gives $t_3^{ps}(13) \leq 24$; all 267 classes of PSTS$(13, 24)$ are non-orientable (the SAT instances are unsatisfiable), while exactly one of the 9,348 classes of PSTS$(13, 23)$ is orientable. Hence $t_3^{ps}(13) = 23$. The orientable class has trivial automorphism group and every point on 5 blocks and 2 ordinary lines; we verified that the oriented matroid returned by the solver satisfies all rank-3 chirotope axioms (the three-term Grassmann–Plücker relations, their contractions, and the matroid basis-exchange axiom on its support). It is non-realizable over $\mathbb{R}$ because it contains an open Pappus configuration, which is exactly why $t_3(13) = 22 < 23$.

For $n = 14$: the counting bound gives $t_3^{ps}(14) \leq 28$. One of the classes of PSTS$(14, 27)$ is orientable (verified against all chirotope axioms), so $t_3^{ps}(14) \geq 27$; and all 787 classes of PSTS$(14, 28)$ (each necessarily 6-regular, with a perfect matching as leave) are non-orientable. Hence $t_3^{ps}(14) = 27$.

For $n = 15$ the same counting gives $31 \leq t_3^{ps}(15) \leq 32$: the pseudoline maximum satisfies exactly the dichotomy that the straight-line problem posed, and $t_3^{ps}(15) = 32$ would hold iff

one of the 7,955,290 classes of PSTS(15, 32) is orientable. We leave this determination open. A global satisfiability search for such an orientation did not terminate within our budget, and an orientability test on more than 3,800 of the classes, sampled across all 13 leave types, found none. The fraction of orientable systems drops sharply with n (one class in 9,348 at $(13, 23)$, one in 314,092 at $(14, 27)$), so a decisive sample would have to be very large; we regard $t_3^{ps}(15)$ as the natural companion open problem to Theorem 1.1.

6 Verification

Enumeration. (i) With the symmetry-breaking rule (R) switched off, the search for $n = 13$ (all leave types for 23 and 24 blocks) produces exactly the same sets of canonical forms, from roughly 500 times as many search nodes. (ii) With an empty leave the program reproduces the classical numbers of Steiner triple systems: 1 of order 7, 1 of order 9 and 2 of order 13. (iii) Five of the 13 leaves for $n = 15$ (types 1, 3, 4, 7 and 10 in Table 2) are edge-disjoint unions of three triangles. Adding the three triangles to a PSTS(15, 32) with such a leave gives a Steiner triple system STS(15); conversely every STS(15) together with a choice of three blocks with the appropriate intersection pattern gives such a system. The completions of the 304,039 enumerated classes with decomposable leaves are exactly 80 pairwise non-isomorphic STS(15)—the classical number [12]—and, counting for each of these 80 systems the isomorphism classes of (system, marked triple of blocks) with nauty, we recover exactly the five per-leave counts 19,711, 116,605, 24,734, 116,876 and 26,113 of Table 2. (iv) An independent enumerator, written separately in Python with a different search strategy (exact cover branching on the pair with fewest candidate triangles, isomorph rejection by orbit–stabiliser computations in the automorphism group of the leave, and its own canonical forms via pynauty) reproduces, class for class, all 13 leave types for $n = 13$ with 23 blocks, the three types with 24 blocks, all seven types for $n = 14$ with 27 blocks, the type with 28 blocks, and the types 1, 4 and 10 for $n = 15$ (19,711, 24,734 and 26,113 classes); it also re-derives the lists of leave types by its own orderly generation.

Realizability. (iv) For $n = 13$ the algebraic test was run on all 9,348 systems with 23 blocks *without* the closure filter: every system is rejected (7496 degenerate, 1852 without complex solutions), so the closure filter never rejected a system that the algebra would have accepted. The same experiment on random samples of closure-rejected systems for $n = 15$ gave the same outcome (1000 of 1000 rejected). (v) The oriented-matroid test of Section 4.3 was run on 84 randomly chosen systems for $n = 15$ spanning all verdict classes, and on all systems for $n = 13$ with 24 blocks; every instance is unsatisfiable. Thus for these systems even a pseudoline arrangement with the prescribed collinearities does not exist. (vi) As positive controls, the Sylvester configurations on $\mathbb{Z}/9$, $\mathbb{Z}/13$ and $\mathbb{Z}/15$ (which are realizable on a cubic), the Pappus and the Desargues configurations pass the closure filter, are found to have solutions of the expected dimension by the algebraic test, and are satisfiable in the oriented-matroid test, while the Fano plane is rejected by both.

7 Concluding remarks

The next open case is $n = 17$, where $t_3(17) \in \{40, 41, 42\}$ and the analogous reduction leads to PSTS(17, 42) (leaves with 10 edges, 26 types) and PSTS(17, 41) (leaves with 13 edges, 230 types). Since $17 \equiv 5 \pmod 6$ and a maximum packing of K_{17} by triangles has 44 blocks, these

systems are two or three blocks short of maximum packings, and their numbers are expected to exceed the 15-point count by several orders of magnitude; a direct extension of the present method appears to require geometric pruning during the enumeration rather than afterwards. Deciding the pseudoline analogue $t_3^{ps}(15)$ (Section 5.1) would require running the orientability test on all 7,955,290 classes, or a global orientability search that terminates; both are beyond the budget of this paper.

8 Data and code

The programs (`leaves.py`, `psts.c`, `closure_fast.py`, `realize.py`, `survivors.py`, `omcnf.c`, `sts15_check.py`, `reduction.py`), the lists of all isomorphism classes of partial triple systems, and the verdict files with witnesses (for closure rejections: the embedded configuration) accompany this paper. Software used: nauty 2.8, Singular 4.3.2, CaDiCaL 1.7, Python 3.11, GCC 13. Total computing time: about 90 core-hours on two cores.

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